MIE 402 · ME LAB II

Lab 4: free vibration of a 1-DOF system

From a sensor-cart record to mass, stiffness, natural frequency, and damping

Fall 2026 · Lab 4 theory · October 19

A small cart teaches the same identification logic used in full-scale vibration testing.



NASA/GSFC image · TESS vibration test

Why this experiment matters



Vehicle suspension

A bump excites the system. Frequency and decay tell engineers whether the suspension will settle safely.



Spacecraft qualification

Modal and vibration tests identify resonances before launch loads expose a weak design.

Lab 4 uses the same measurement logic at a smaller scale: disturb, record the free response, identify frequency and damping, then compare with the model.

The laboratory question and the evidence

One record contains two independent clues

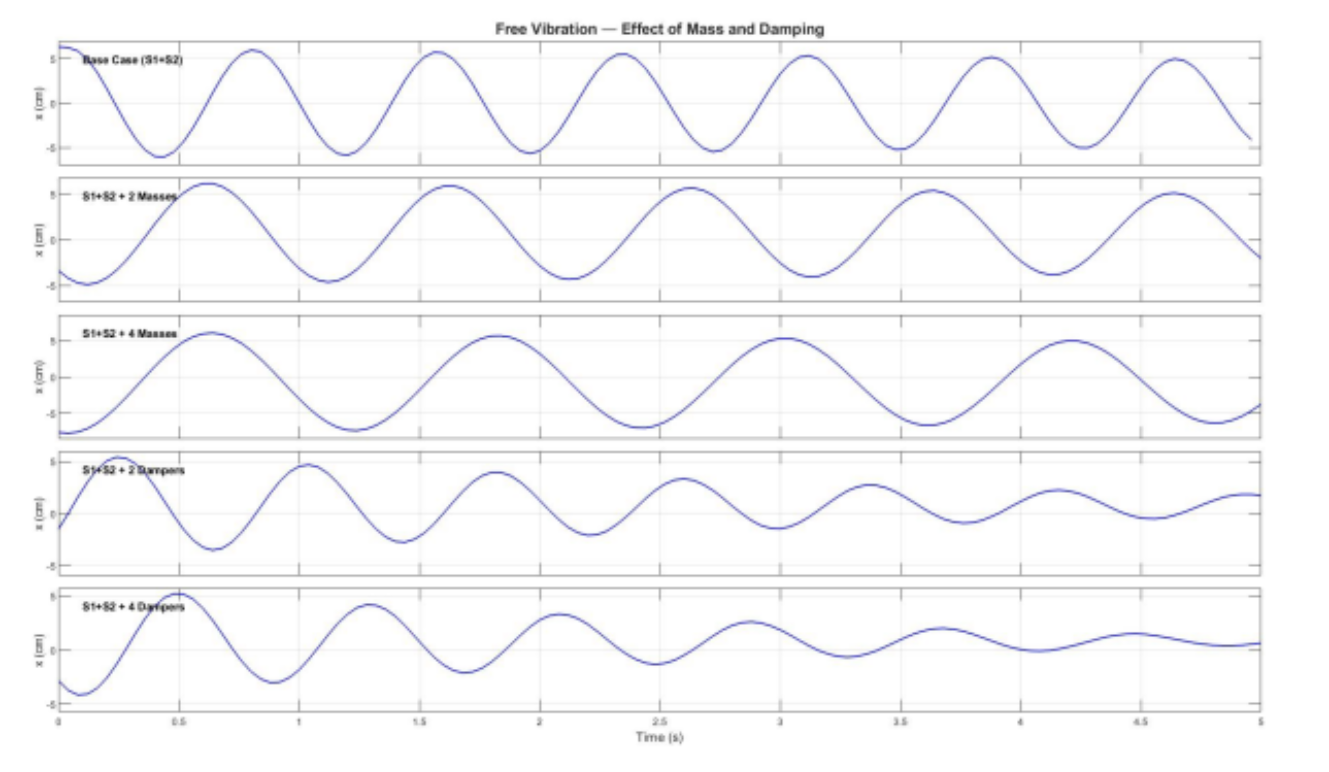
Peak spacing

sets the oscillation period and damped natural frequency.

Peak decay

sets the logarithmic decrement and damping ratio.

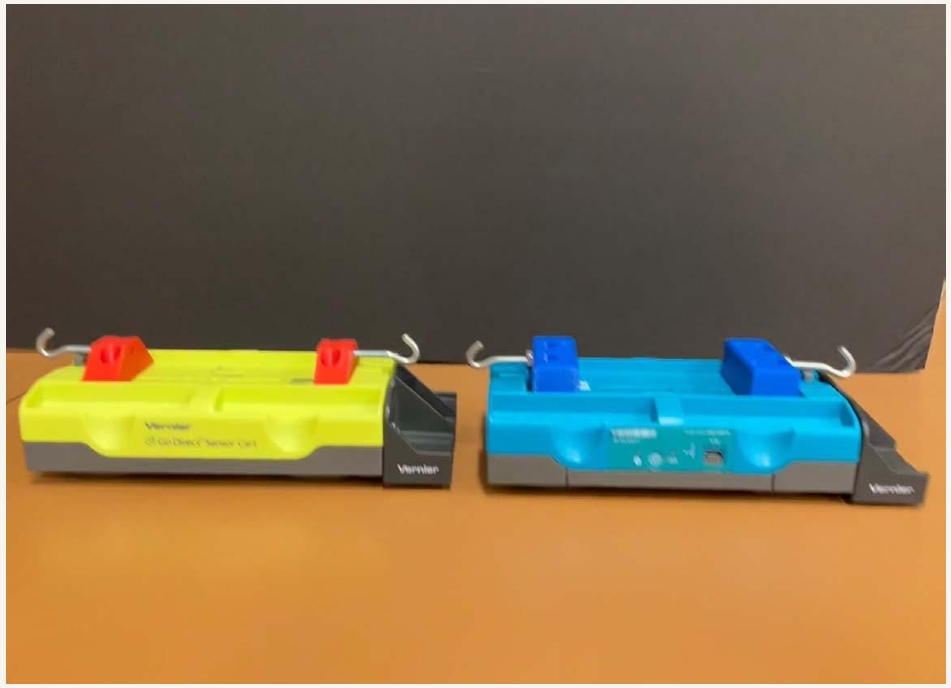
The seven cases change mass, damping, or spring pair one at a time so the cause of each change remains visible.

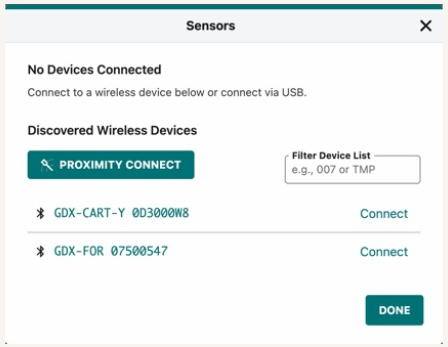


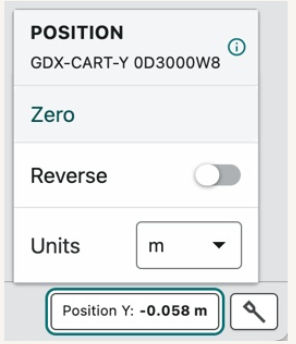
Representative anonymized class data · Spring 2026

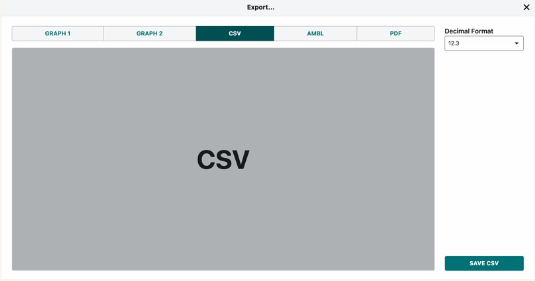
The engineering question: which physical change moved the frequency, and which change shortened the ring-down?

The apparatus records position, not “damping”









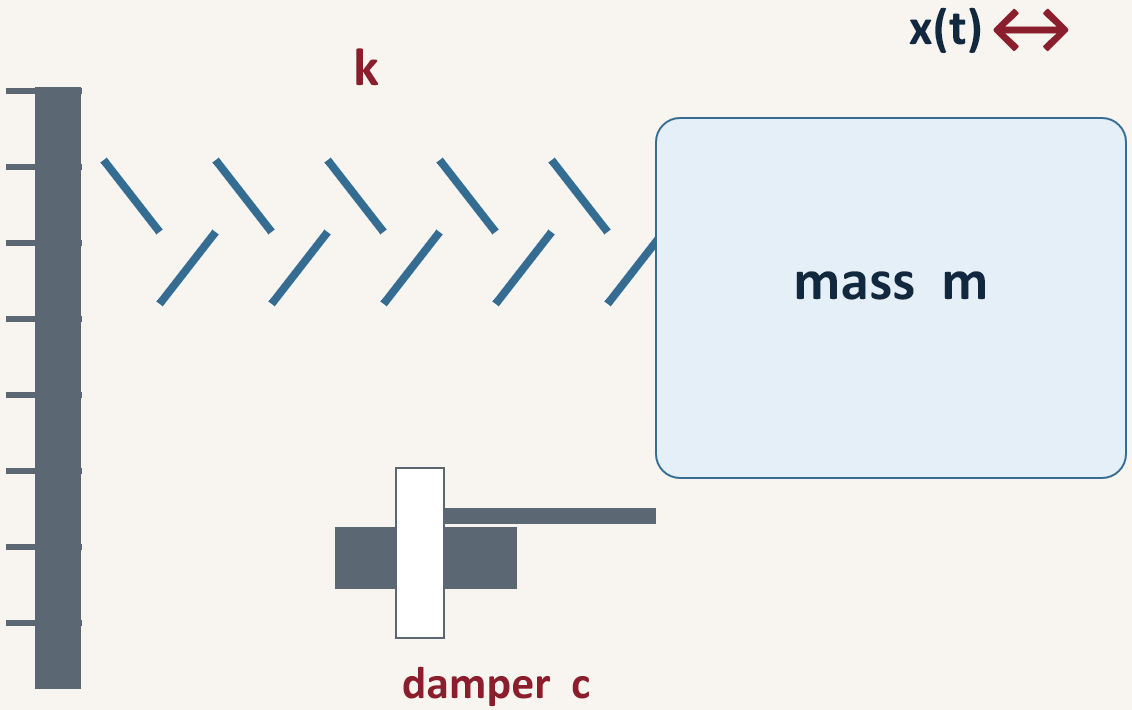
1 wheel encoder records x(t)

2 Graphical Analysis stores time and position

3 CSV carries the raw evidence into MATLAB

The sensor never reports “damping.” Students infer damping from how the measured peaks decrease.

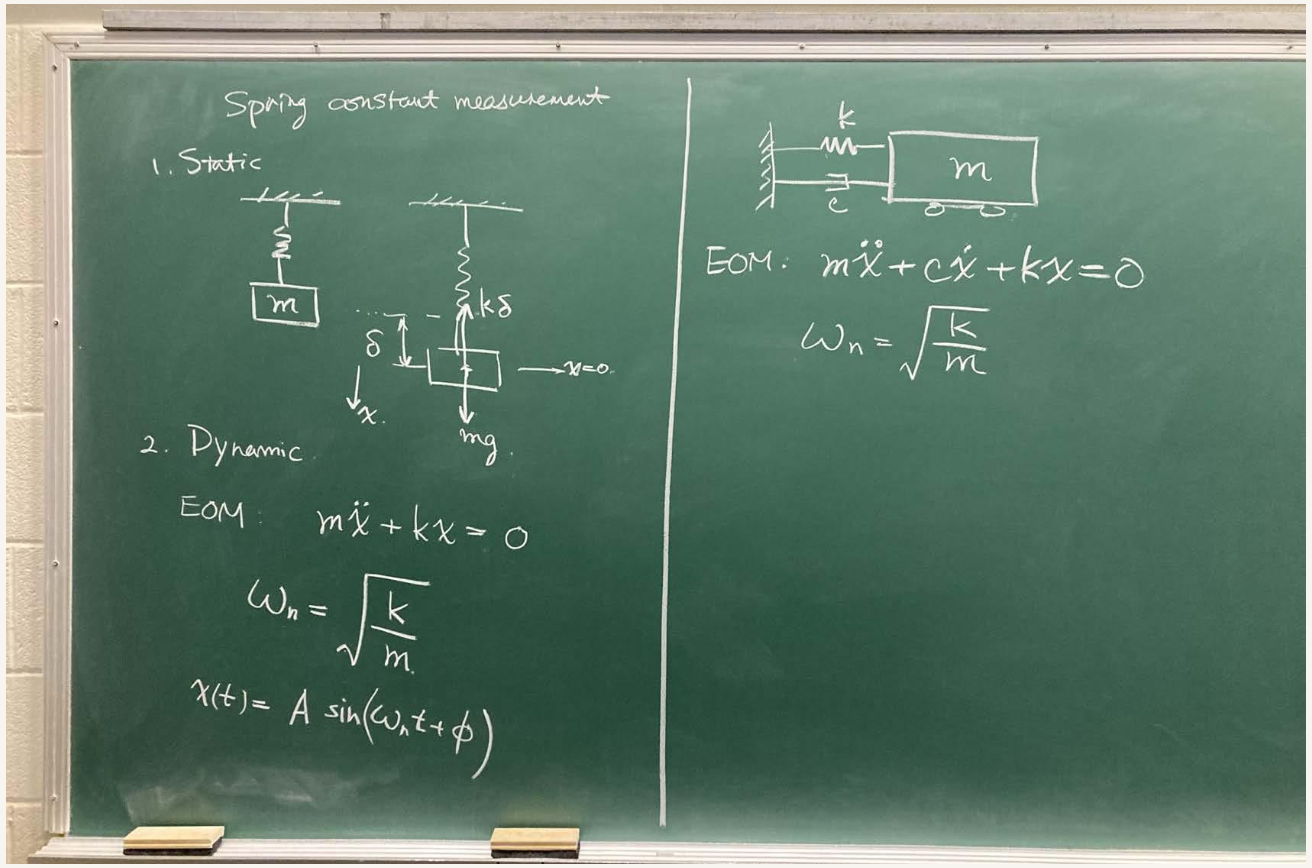
The hardware becomes a spring–mass–damper model



$$m\,\ddot{x} + c\,\dot{x} + k\,x = 0$$

- x(t): cart displacement from equilibrium [m]
- m: total moving mass [kg]
- c: viscous-equivalent damping coefficient [N·s/m]
- k: equivalent stiffness of both springs [N/m]

A free-body diagram gives the equation of motion



Prior MIE 402 classroom derivation

Start from forces, then simplify

Spring force $F_s = -kx$

Damping force $Fd = -c\,\dot{x}$

$$m\,\ddot{x} + c\,\dot{x} + kx = 0$$

The equation describes motion after release, when no external input remains.

Each parameter changes a different feature of the record

Mass m

More moving mass stretches the period.
The FFT peak shifts left.

Stiffness k

A stiffer spring pair shortens the period.
The FFT peak shifts right.

Damping c

More eddy-current damping reduces peak height faster. Frequency changes only slightly.

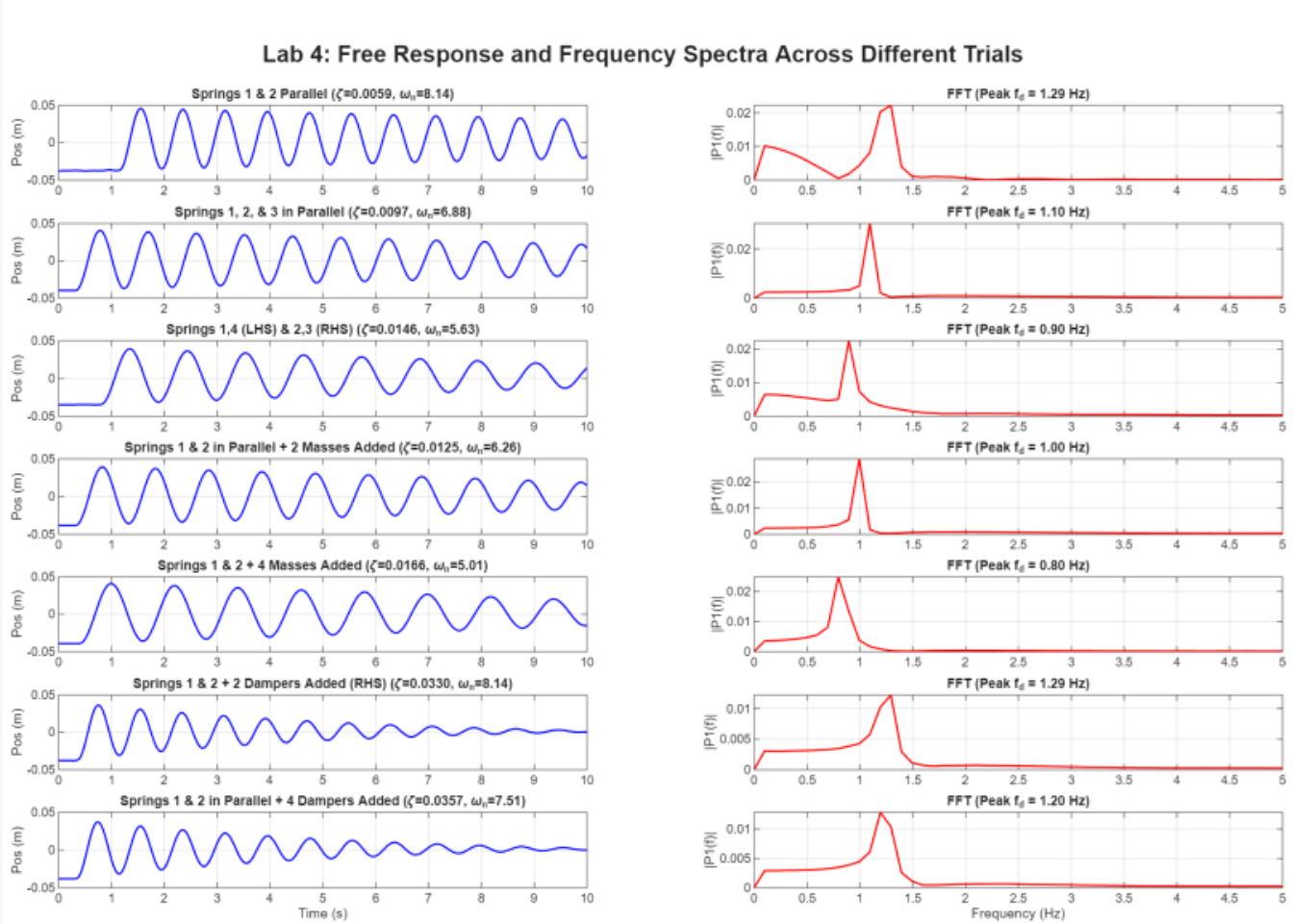


Figure 1: Lab 4 - Free Response and Frequency Spectra Across Different Trials

Representative anonymized class result · Spring 2026

Without damping, the cart exchanges energy forever

$$m\ddot{x} + kx = 0$$

$$\omega_n = \sqrt{(k/m)} \quad [\text{rad/s}]$$

$$x(t) = A\cos(\omega_n t - \phi)$$

$$f_n = \omega_n/(2\pi) \quad [\text{Hz}]$$

ω_n is the undamped natural angular frequency. f_n counts cycles per second. A and ϕ come from the initial displacement and velocity.

Dimensional check: k/m has units s^{-2}, so $\sqrt{(k/m)}$ has units rad/s.

Natural frequency is a property of the system

Double the mass

f_n becomes $1/\sqrt{2} = 0.707$ times the original

Double the stiffness

f_n becomes $\sqrt{2} = 1.414$ times the original

Double m and k

f_n stays unchanged

$$f_n \propto \sqrt{k} \quad \text{and} \quad f_n \propto 1/\sqrt{m}$$

Energy explains the repeated motion

$$E(t) = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2$$

turning point

$x = \pm A$

spring energy maximum
speed = 0

equilibrium

$x = 0$

spring energy minimum
speed maximum

With c = 0, total mechanical energy stays constant. Real damping removes energy on every cycle.

Damping removes energy fastest when the cart moves fastest

$$F_d = -c\,\dot{x}$$

$$P_d = F_d\,\dot{x} = -c\,\dot{x}^2$$

- The damping force reverses when velocity reverses.
- The damper does no work at a turning point because $\dot{x} = 0$.
- The magnitude of energy loss is largest near equilibrium, where speed is largest.
- The magnetic dampers approximate viscous damping over the operating range.

We model several loss mechanisms with one equivalent coefficient c. The model is useful only if the measured envelope resembles an exponential.

Damping ratio classifies the response

$$\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0$$

$$\zeta = c / (2\sqrt{km})$$

$0 < \zeta < 1$

underdamped

oscillates and decays

$\zeta = 1$

critical

fastest nonoscillatory return

$\zeta > 1$

overdamped

slow nonoscillatory return

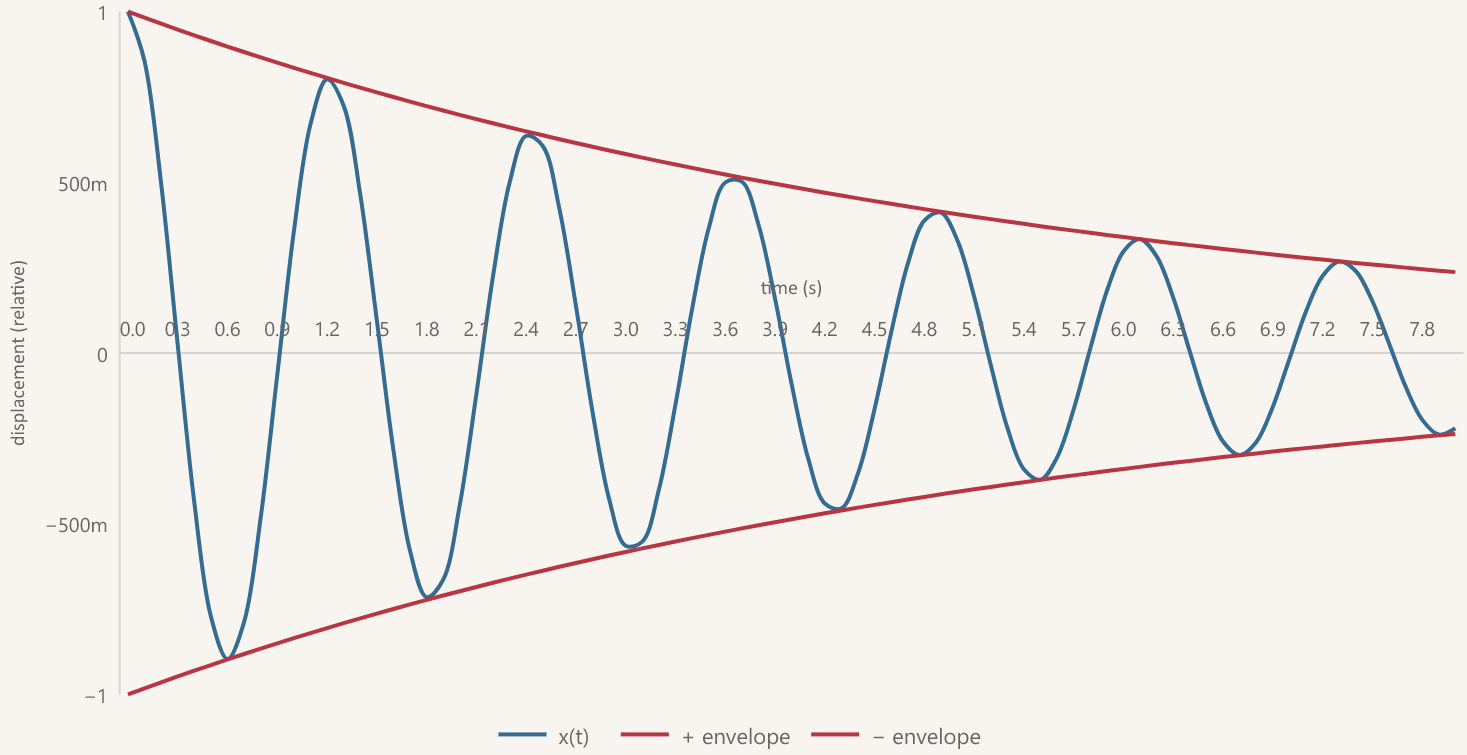
Lab 4 operates in the underdamped regime

$$x(t) = A \exp(-\zeta \omega_n t) \cos(\omega_d t - \phi)$$

$$\omega_d = \omega_n \sqrt{(1 - \zeta^2)}$$

- A: response amplitude set by the initial conditions [m]
- ϕ: phase set by initial displacement and velocity [rad]
- ω_d: damped natural angular frequency [rad/s]
- $\exp(-\zeta \omega_n t)$: exponential envelope that controls decay

The peaks trace an exponential envelope



$$\text{envelope} = \pm A \exp(-\zeta \omega_n t)$$

Peak spacing gives the period. Peak-height reduction gives damping.

Use peaks with the same sign, or use absolute peak magnitudes consistently.

Logarithmic decrement turns decay into a ratio

$$\delta = (1/N)\ln(|x_i| / |x_{i+n}|)$$

$$\zeta = \delta / \sqrt{(4\pi^2 + \delta^2)}$$

- x_i and x_(i+N) are peak amplitudes of the same sign.
- N is the number of complete cycles between the selected peaks.
- Using several cycles reduces sensitivity to cursor placement and noise.
- A line fit to ln|peak| versus time provides a stronger check.

28–30 min

Peak timing gives damped and undamped frequency

$$T_d = (t_{i+n} - t_i)/N$$

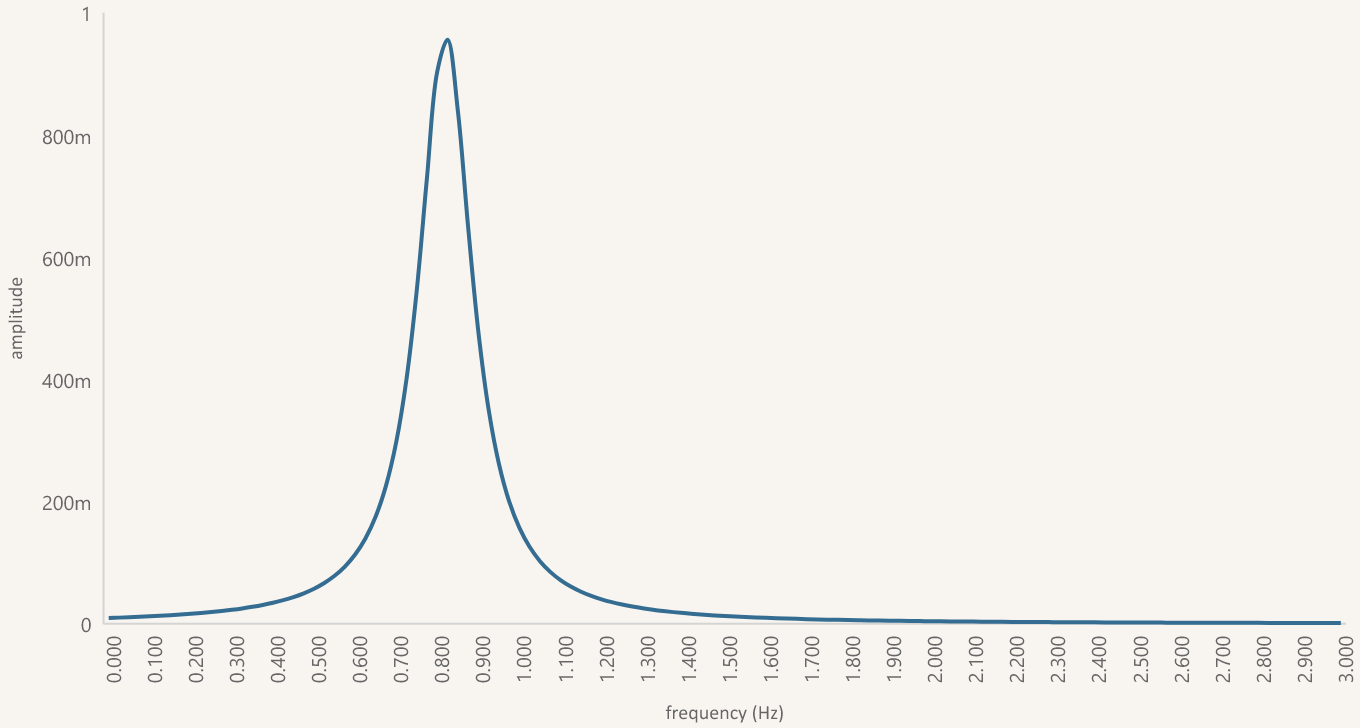
$$f_d = 1/T_d$$

$$\omega_d = 2\pi f_d$$

$$\omega_n = \omega_d / \sqrt{(1 - \zeta^2)}$$

For light damping, f_d and f_n are close, but they are not identical. Report which one you measured and which one you calculated.

The FFT finds the frequency that best matches the record



largest spectral peak

$$f_{peak} \approx fd$$

The cart repeats most strongly at its damped natural frequency, so the correlation with that sinusoid becomes largest.

Time-domain and FFT estimates must agree

peak spacing

$f_{d,time}$

direct and intuitive

FFT peak

f_{peak}

fast visual check

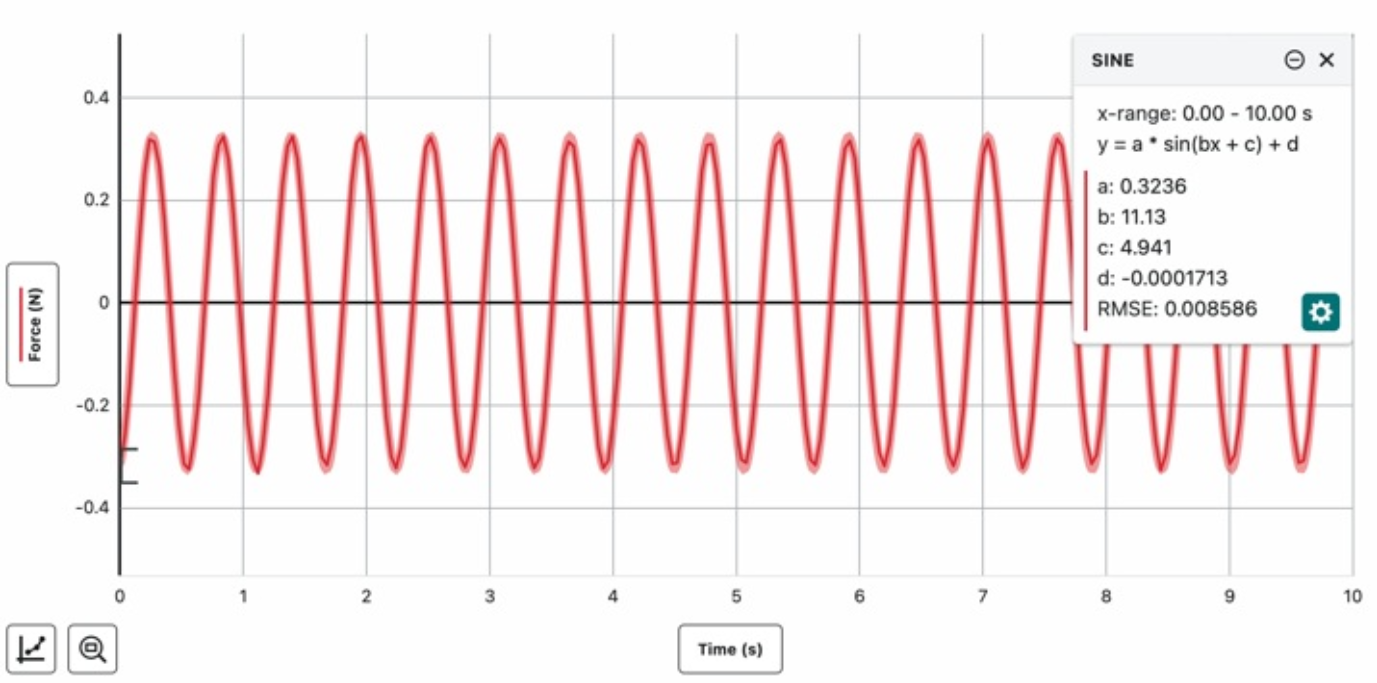
model fit

f_d, ζ

uses the entire record

If the estimates disagree, inspect sampling, drift, peak selection, units, and whether the record contains more than one motion.

Spring stiffness comes from a separate free-vibration test



Actual Graphical Analysis fit from a prior MIE 402 setup

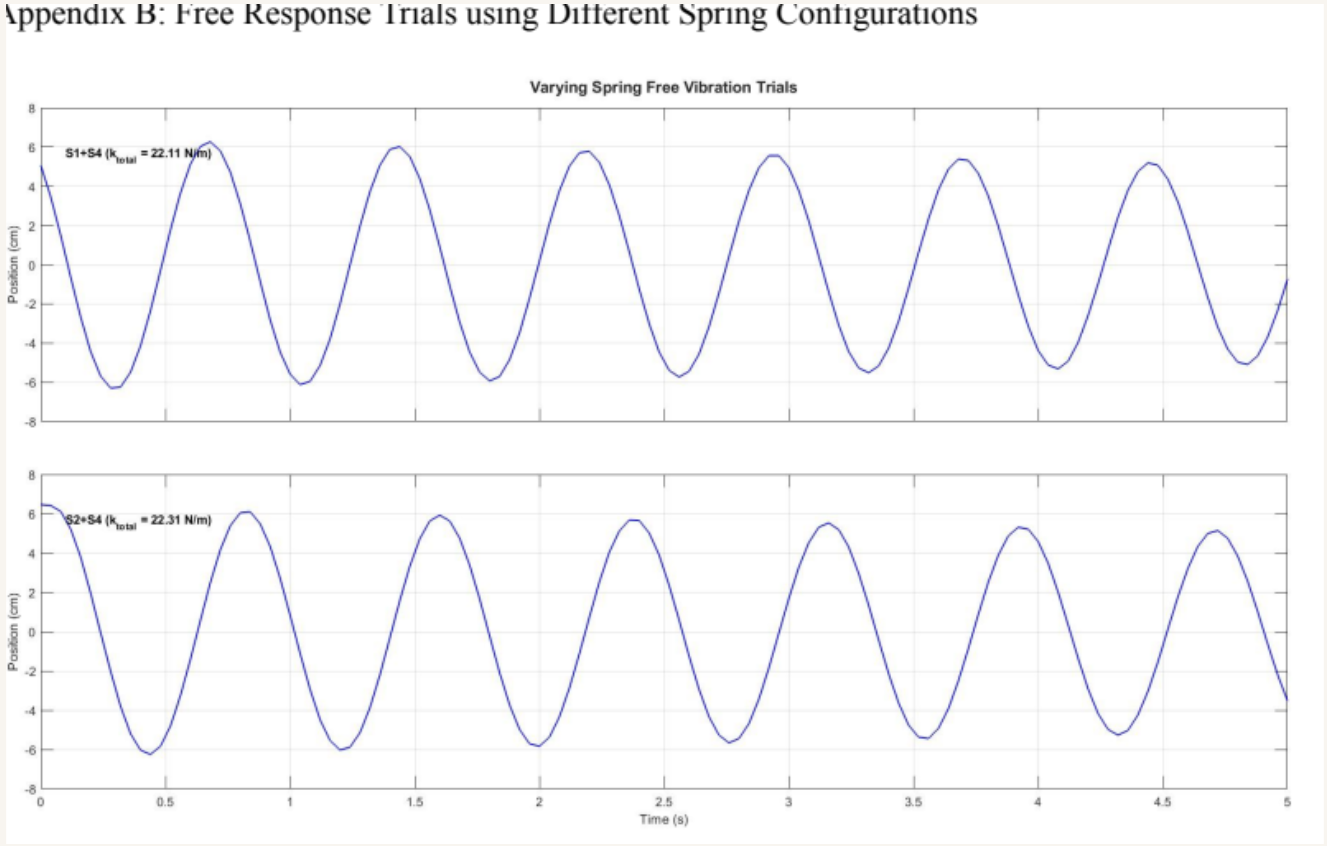
$$k \approx m_h (2\pi f_s)^2$$

What the fit reports

- a: force amplitude
- b: angular frequency in rad/s
- c: phase
- d: force offset

Use b directly: $k \approx m_h b^2$. Include every moving attachment in m_h.

Two springs act together on the cart



Representative anonymized class data · Spring 2026

A small displacement stretches one spring and compresses the other. Both forces oppose the displacement.

$$k = kL + kR$$

$$f_n = (1/2\pi)\sqrt{(k/m)}$$

The prior runs at left show similar but measurably different periods for two spring combinations.

What a complete seven-case data set can show

Configuration	Theoretical ω_n (rad/s)	Measured ω_n (rad/s)	Damping Ratio (ζ)	Error
Base Case	7.92	8.18	0.0057	3.3%
Springs 1&2, 2 Dampers	7.75	8.05	0.0286	4.0%
Springs 1&2, 4 Dampers	7.59	7.86	0.0396	3.6%
Springs 1&2, 2 Masses	6.03	6.28	0.0075	4.2%
Springs 1&2, 4 Masses	5.06	5.30	0.0104	4.6%
Springs 1&4	8.00	8.36	0.0060	4.4%
Springs 2&4	8.04	8.10	0.0054	0.7%

Results from a prior section

Mass cases — **5.30–6.28 rad/s**

Base case — **8.18 rad/s**

Spring cases — **8.10–8.36 rad/s**

Damper cases — **ζ rose to 0.0396**

The useful result is the trend and the theory–experiment comparison, not a single “correct” number.

Collection settings must preserve the motion

sample rate — **≤ 25 Hz**

enough samples per cycle

record length — **until nearly at rest**

captures the decay envelope

initial displacement — **measure and record**

release without a push

position zero — **set before each run**

prevents avoidable offsets

42–44 min

A reliable run follows the same sequence every time

1. Level track and confirm clear travel
2. Pair the cart by serial number
3. Zero position at equilibrium
4. Displace a measured distance
5. **Click Collect, then release without push**
6. Record until motion nearly stops
7. Export CSV with a descriptive file name
8. Plot immediately and repeat if needed

The analysis code produces four pieces of evidence

1

clean x(t)

remove offset and inspect drift

2

peak table

times, amplitudes, and cycle spacing

3

decay estimate

δ, ζ, and fitted envelope

4

frequency check

time-domain f_d and FFT peak

Keep raw CSV files unchanged. The code writes figures and a summary table to a separate results folder.

Quality checks catch most bad runs before students leave

Accept the run when…

- several complete cycles
- smooth exponential-like envelope
- cart remains on track
- no clipped or missing samples
- metadata matches file name

Repeat the run when…

- push at release
- spring goes slack or overstretches
- cart or cable rubs
- track not level
- wrong mass or spring recorded

Fully analyze one case and show the result to the GTA before leaving.

48–49 min

Pre-Lab 4 prepares the decisions made at the bench

- Identify the industrial role of free-response testing.
- Calculate ω_n, ζ, and ω_d for a new spring–mass–damper case.
- Predict how mass, stiffness, and damping change the record.
- Run provided code for an analytical response and a numerical solution.
- Use provided peak-analysis code to estimate damping and frequency.
- Submit one Word or PDF file with answers and required figures before your own Lab 4 section begins.

No Python or MATLAB programming is required. Students run supplied code, interpret outputs, and explain the physics.

Lab 4 evidence feeds the combined Labs 4–5 report

Model

define m, k_eq, c, ω_n, ζ, and assumptions

Method

describe calibration, cases, sampling, and release

Results

time traces, spectra, parameter table, uncertainties

Discussion

compare trends with theory and diagnose deviations

The central claim should answer: do the measured changes in natural frequency and damping follow the spring–mass–damper model?